

Reasoning Primitives in Hybrid and Non-Hybrid LLMs

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Abstract

We study reasoning through two primitives—recall and state-tracking—and ask whether hybrid architectures combining attention-based retrieval with recurrent state updates outperform attention-only models on tasks requiring both. Using matched OLMo3 transformer and hybrid models in instruction-tuned and reasoning-augmented variants, we evaluate five controlled task families. Reasoning augmentation provides the largest overall improvement, but its benefit has a ceiling: at high difficulty, the hybrid Think model remains robust on tasks with sequentially chained state updates, while the transformer Think model is more robust on tasks with flat multi-hop retrieval. Parse rate figures reveal a further dimension—the transformer Think model’s collapse on the collision task is a generation breakdown, not merely an accuracy failure, while the hybrid Think model on Dyck maintains accuracy even as its parse rate degrades. We present these findings as suggestive rather than conclusive.

1 Introduction

Chain-of-Thought reasoning substantially improves LLM performance on complex tasks (Wei et al., 2022), and this picture has been extended across prompting strategies (Yao et al., 2023), architectures (Geiping et al., 2025), and training regimes (Wang et al., 2025; Yang et al., 2025; Bakouch et al., 2025; Zeng et al., 2026). The harder question is why. Levy et al. (Levy et al., 2025) propose the *State over Tokens* (SoT) framework: because internal activations do not persist across generation steps, intermediate tokens serve as the only durable carrier of state. Reasoning tokens improve performance not by making the model explain itself, but by letting it externalize partial results and transform bounded computations into cumulative ones (Kambhampati et al., 2025; Liu et al., 2025).

This framing makes architecture central. Transformers excel at content-based retrieval but face

well-known scaling constraints (Fichtl et al., 2025; Li et al., 2024); recurrent and state-space architectures improve efficiency but sacrifice precise addressable retrieval (Yang et al., 2024b,a; Dao and Gu, 2024). Hybrid architectures combine both (Merrill et al., 2026). We ask: under what conditions do architectural differences become visible when reasoning is decomposed into its constituent computational demands?

2 Primitives, Tasks, and Methods

Reasoning primitives. Following Merrill et al. (Merrill et al., 2026), we decompose reasoning into *state-tracking*—maintaining and updating structured variables across sequential transformations—and *recall*—retrieving specific information from a large context with high precision. When combined into **state-based recall**, the retrieval address is itself determined by a sequence of state transformations, making the task irreducible to either primitive alone (see Appendix B for canonical examples).

Models. We compare OLMo3-7B and OLMo3-Hybrid-7B (Merrill et al., 2026), a matched pair sharing the same data, optimization schedule, and training recipe, in both instruction-tuned and reasoning-augmented variants: OLMo-3-7B-{Instruct,Think} and OLMo-Hybrid-{Instruct,Think}-SFT-7B. Inference used vLLM (Kwon et al., 2023); Think variants: temperature 0.0, max 6,000 tokens; Instruct variants: temperature 0.0, max 40 tokens.

Evaluation. All tasks are free-form generation scored by exact matching. Difficulty is parameterised by m (retrieval complexity) and n (state maintenance complexity), with 1000 instances per (m,n) bin except for the DYCK task which are only 500 instances per (m,n) . All tasks require JSON output {"answer": "A|B|C|D"}. We report *accuracy* and *parse rate* (fraction of parseable responses) separately, since the two can dissociate: at high difficulty, Think models can fail to produce any structured output—a generation breakdown distinct from an incorrect answer.

Tasks. We evaluate five procedurally generated families (prompt examples in Appendix C). **State-**

based **Astro Recall** (*astro*): m real exoplanet rows in a Markdown table; variables initialized to orbital period values undergo n simultaneous swaps; the model retrieves the planet name for the queried variable. **Collision Simulator** (*collisions*): m particles with distinct velocities undergo n sequential elastic collisions (velocity exchange); each collision conditions all subsequent ones, creating compounding sequential dependencies. **OLMo Original** (*olmo_original*): minimal state-based recall from Merrill et al. (2026)—a bit array of length m , m pointer variables, n simultaneous swaps, evaluate `bits[a]`; replication baseline. **Dyck Language** (*dyck*): bracket sequence with one masked closer; model maintains a stack to identify the correct token; m = stack depth, n = sequence length; no retrieval component. **DAG Arithmetic** (*dag*): m input variables, n computation layers introducing new variables via addition/subtraction with arbitrary cross-layer references (Motwani et al., 2026); flat multi-hop retrieval over a growing registry.

3 Results

3.1 Collision Simulator: Hybrid Advantage

Under instruction-only prompting, both architectures hover between 0.27–0.40, near chance. The Think variants diverge sharply beyond (16,16): OLMo-3-7B-Think falls to 0.49 at (32,32) and collapses to near zero at (64,64); OLMo-Hybrid-Think-SFT-7B retains 0.94 and 0.75 at the same settings (Figure 1). Figure 2 presents a contrasting trend, showing that the parse rate of OLMo-Hybrid-Think-SFT-7B remains consistently below that of OLMo-Think-SFT-7B. However, this suggests that although OLMo-Hybrid-Think-SFT-7B produces fewer successfully parsed outputs overall, the responses that are successfully parsed are more likely to contain the correct answer. In contrast to the transformer-based Think model, whose degradation is reflected in an increasing inability to generate parseable outputs. This behavior is consistent with the hypothesis that hybrid recurrence stabilizes intermediate state propagation over long sequential reasoning chains, thereby preserving solution quality even when output formatting deteriorates.

3.2 DAG Arithmetic: Transformer Advantage

Both Think models reach perfect accuracy through (8,8). Beyond (16,16) the pattern reverses: OLMo-3-7B-Think retains 0.56 at (32,32) while OLMo-Hybrid-Think-SFT-7B falls to 0.30 (Figure 3). Crucially, parse rates remain high for both models (Appendix Figure 6), confirming that this

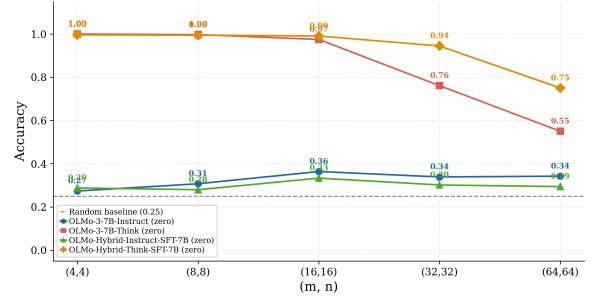


Figure 1: Collision Simulator accuracy. Hybrid Think retains high accuracy where transformer Think collapses.

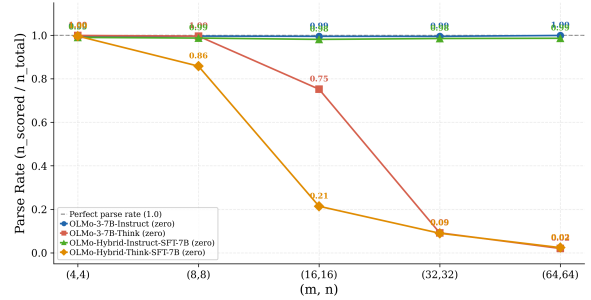


Figure 2: Collision Simulator parse rate. Transformer Think and hybrid Think both exhibit declining parse rates, though hybrid Think shows greater difficulty adhering to the required output format.

gap reflects genuine computational failure rather than a generation breakdown. The flat, multi-hop retrieval structure of DAG arithmetic—where any prior variable may be referenced at any later layer—appears to favour the transformer’s content-addressable attention over the hybrid’s recurrent state updates.

3.3 Remaining Tasks

Astro Recall. Both Think models achieve ≈ 0.95 – 0.97 at (4,4) but collapse to 0.00/0.24 by (32,32); Instruct variants hover near chance throughout. The joint demand of simultaneous swaps and large-table lookup exceeds what any model sustains beyond moderate difficulty (Appendix Figures 4, 5).

Dyck Language. OLMo-Hybrid-Think-SFT-7B achieves the highest accuracy in the study (peak 0.79, never below 0.38 at any setting); OLMo-3-7B-Think is substantially more volatile (peak 0.67, dip to 0.22). Both Instruct variants hover near the 0.33 baseline. Notably, the hybrid Think model’s parse rate degrades at high difficulty while its accuracy remains comparatively high—suggesting its recurrent components support correct internal computation even when output structure fails (Appendix Figures 7, 8).

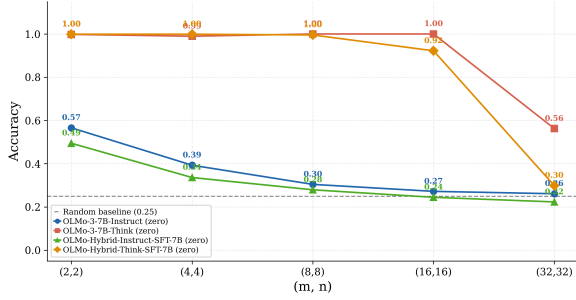


Figure 3: DAG Arithmetic accuracy. Transformer Think is more robust at high difficulty, reversing the collision pattern.

OLMo Original. Both Think models begin near-perfect (≈ 0.91 – 0.92) and degrade in parallel toward chance—no consistent architectural advantage at the Think level. At the Instruct level, an unexpected reversal: transformer Instruct outperforms hybrid Instruct (0.43 – 0.53 vs. 0.32 – 0.43) (Appendix Figures 9, 10).

4 Discussion and Conclusion

Two patterns are consistent across all five tasks. First, **reasoning augmentation is the dominant factor**: Think variants substantially outperform Instruct variants at low-to-moderate difficulty, consistent with the SoT account that externalizing state into tokens enables deeper cumulative computation. Second, **this mechanism has a ceiling**, and what fills the gap depends on task structure.

The collision and DAG arithmetic results establish this most precisely. On collisions—strictly chained updates where each step overwrites prior state—the hybrid Think model’s recurrent components supplement token-externalized state when reasoning alone fails, and sustain generation where the transformer collapses entirely. On DAG arithmetic—flat multi-hop retrieval over a growing variable registry—the transformer’s content-addressable attention proves more robust, and both models maintain high parse rates, confirming genuine computational rather than generation failure. The Dyck parse-rate/accuracy dissociation adds a further nuance: the hybrid model’s recurrent components may support correct computation at a level not fully captured by structured-output scoring.

These results suggest **architectural inductive biases are task-specific affordances**: chained sequential updates favour hybrid recurrence; flat multi-hop retrieval favours transformer attention. The clearest methodological takeaway is that benchmarks for architectural evaluation should target regimes where reasoning traces are helpful but insufficient, stressing both chained and flat re-

trieval structures—and should report parse rate alongside accuracy to distinguish computational failures from generation breakdowns.

5 Limitations

Model family scope. All experiments use a single matched OLMo3 pair. Whether findings replicate across other hybrid architectures (Mamba, RWKV), other transformer families, or other scales remains open.

Single parameter scale. All models are evaluated at 7B. Relative advantages and difficulty thresholds are likely scale-dependent.

Task coverage. The five families probe recall, state-tracking, and their composition in synthetic settings. Performance here does not license claims about general reasoning ability.

Zero-shot only. It is not possible to determine how much of the observed gap reflects architectural differences versus distributional mismatch. Fine-tuning experiments would allow a cleaner test.

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332 A Additional Result Figures

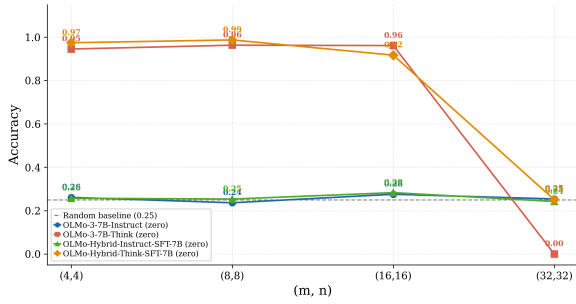


Figure 4: Astro Recall accuracy. Both Think models collapse by (32, 32).

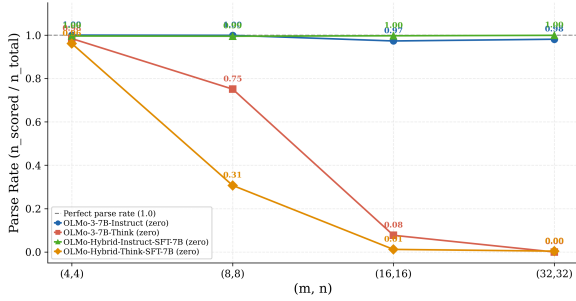


Figure 5: Astro Recall parse rate. Degrades alongside accuracy for both Think models.

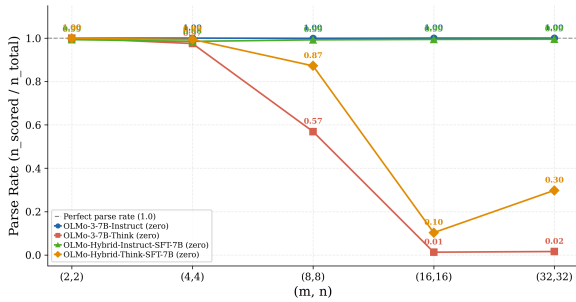


Figure 6: DAG Arithmetic parse rate. Both Think models remain parseable throughout; accuracy gap reflects computational failure.

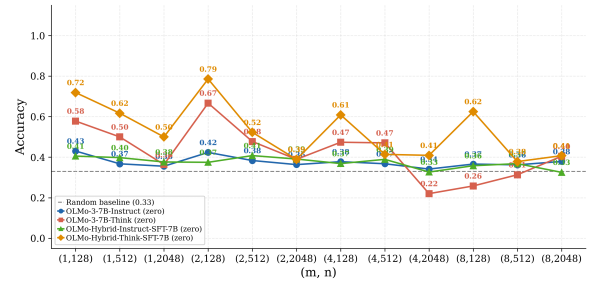


Figure 7: Dyck accuracy. Hybrid Think achieves the highest scores in the study; transformer Think is more volatile.

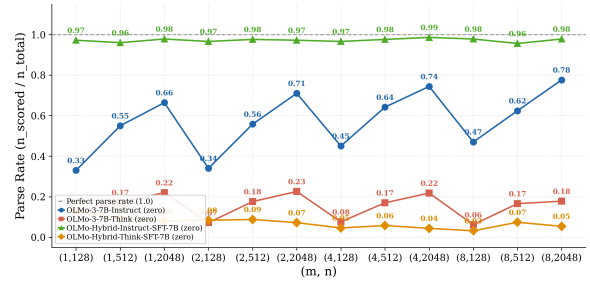


Figure 8: Dyck parse rate. Hybrid Think parse rate degrades at high difficulty while accuracy remains high—recurrent computation persists even when output structure fails.

```
bits = [0, 1, 0, 0, ...]
a = 42
assert bits[a] == _
```

338

Example 3. *Example C. [State-Based Recall]*

339

```
bits = [0, 1, 0, 0, ...]
a, b, c, d, e = 36, 23, 12, 2, 56
a, c = c, e
...
assert bits[a] == _
```

340

C Task Prompt Examples

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C.1 State-based Astro Recall

342

```
| Planet      | Host Star | Orbital Period (days)
| Kepler-423 b | Kepler-423 | 2.684
| TOI-3894 b  | TOI-3894  | 8.798
a, b, c, d = 2.684, 8.798, 5.508, 2.644
Swaps: a,b=b,a then b,c=c,b
Planet with Orbital Period = a?
Options: A) TOI-3894 b B) Kepler-423 b C) HAT-P-18
```

343

333 B Code Examples for Reasoning Primitives

334

335 Example 1. *Example A. [State-Tracking]*

```
a, b, c, d, e = [0, 1, 0, 0, 1]
a, c = c, e
...
assert a == _
```

336

337 Example 2. *Example B. [Recall]*

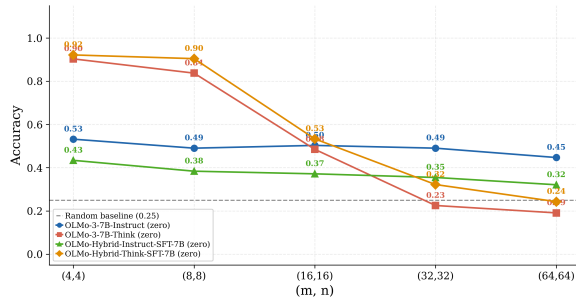


Figure 9: OLMo Original accuracy. Both Think models degrade in parallel; no consistent architectural advantage.

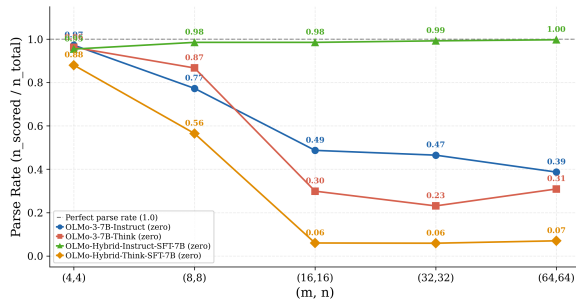


Figure 10: OLMo Original parse rate. Relatively stable; degradation is computational rather than generational.

344 C.2 Collision Simulator

```
Rule: equal-mass elastic collision = velocity exchange.
Initial: A=142, B=57, C=831
1. A collides with B  2. B collides with C
Velocity of A? A)142 B)57 C)831 D)204
```

346 C.3 OLMo Original

```
bits=[0,1,1,0]; a,b,c,d=2,0,3,1
a,b=b,a; b,c=c,b; assert bits[a]==_
Options: A)1 B)0 C)2 D)3
```

348 C.4 Dyck Language

```
Expression: ( [ { _ ] ) ( )
Token at position 4? A) ) B) } C) ]
```

350 C.5 DAG Arithmetic

```
a=3, b=7
v1_0=a+b; v1_1=b-2; v2_1=a+v1_0
Value of v2_1? A)10 B)13 C)15 D)7
```